

Multi-Server Java Application Build, Store, and Deployment System

Objective

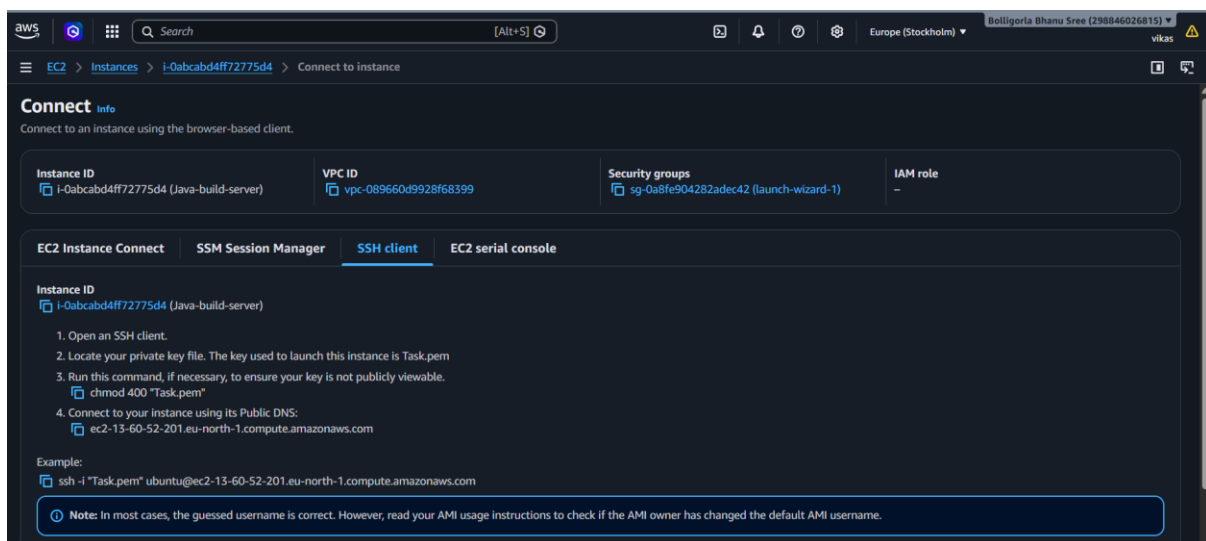
The objective of this task is to set up a multi-server environment for building, storing, and deploying a Java web application using a Build Server, Nexus Repository Server, and Deploy Server.

Servers Used

1. Build Server – Build Java application using Maven
2. Nexus Repository Server – Store WAR artifacts
3. Deploy Server – Deploy application on Apache Tomcat

1. Java Build Server Setup

Step 1: Connect to Build Server



Step 2: Install Java and Maven

```
sudo apt update
```

```
sudo apt install openjdk-17-jdk -y
```

```
sudo apt install maven -y
```

java -version
mvn -version

```
ubuntu@buildvm:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@buildvm:~$
ubuntu@buildvm:~$
```

Step 3: Clone GitHub Repository

git clone https://github.com/mrtechreddy/Java-Web-Calculator-App.git
cd Java-Web-Calculator-App

```
ubuntu@buildvm:~$
ubuntu@buildvm:~$ git clone https://github.com/mrtechreddy/Java-Web-Calculator-App.git
Cloning into 'Java-Web-Calculator-App'...
remote: Enumerating objects: 229, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (10/10), done.
remote: Total 229 (delta 5), reused 2 (delta 2), pack-reused 217 (from 5)
Receiving objects: 100% (229/229), 66.03 KiB | 5.50 MiB/s, done.
Resolving deltas: 100% (53/53), done.
ubuntu@buildvm:~$
```

Step 4: Build Application

mvn clean package

```
ubuntu@buildvm:~/Java-Web-Calculator-App$
ubuntu@buildvm:~/Java-Web-Calculator-App$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.web.cal:webapp-add >-----
[INFO] Building WebAppCal Maven Webapp 0.0.3
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war-plugin-3.3.2.pom (9.7 kB at 21 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war-plugin-3.3.2.pom (9.7 kB at 21 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/34/maven-plugins-34.pom (11 kB at 202 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/34/maven-plugins-34.pom (11 kB at 202 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war-plugin-3.3.2.jar (83 kB at 905 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war-plugin-3.3.2.jar (83 kB at 905 kB/s)
[INFO] --- resources:3.3.1:resources (default-resources) @ webapp-add ---
[INFO] skip non existing resourceDirectory /home/ubuntu/Java-Web-Calculator-App/src/main/resources
[INFO] --- compiler:3.11.0:compile (default-compile) @ webapp-add ---
[INFO] Nothing to compile - all classes are up to date
[INFO] --- resources:3.3.1:testResources (default-testResources) @ webapp-add ---
[INFO] skip non existing resourceDirectory /home/ubuntu/Java-Web-Calculator-App/src/test/resources
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ webapp-add ---
[INFO] Nothing to compile - all classes are up to date
[INFO] ---
```

WAR file generated inside target/ directory.

```
ubuntu@buildvm:~$ cd Java-Web-Calculator-App/
ubuntu@buildvm:~/Java-Web-Calculator-App$ ls
jenkinsfile README.md pom.xml src target
ubuntu@buildvm:~/Java-Web-Calculator-App$ wget http://13.60.183.232:8081/repository/maven-releases/com/web/cal/webapp-add/maven-metadata.xml
--2026-05-18 12:36:22-- http://13.60.183.232:8081/repository/maven-releases/com/web/cal/webapp-add/maven-metadata.xml
Connecting to 13.60.183.232:8081... connected.
HTTP request sent, awaiting response... 200 OK
length: 301 [application/xml]
Saving to: 'maven-metadata.xml'
```

Step 5: Configure pom.xml

Add Nexus repository configuration under distributionManagement.

```
<distributionManagement>
  <repository>
    <id>maven-releases</id>
    <url>http://3.80.155.124:8081/repository/maven-releases/</url>
  </repository>
</distributionManagement>

</project>
```

Step 6: Configure Maven settings.xml

Add Nexus credentials in /etc/maven/settings.xml.

```
<!-- Nexus Authentication -->
<servers>

  <server>
<id>maven-releases</id>
<username>admin</username>
<password>admin123</password>
</server>

  <server>
<id>maven-snapshots</id>
<username>admin</username>
<password>admin</password>
</server>

</servers>

<!-- Mirrors -->
<mirrors>
```

Step 7: Deploy Artifact to Nexus

mvn deploy

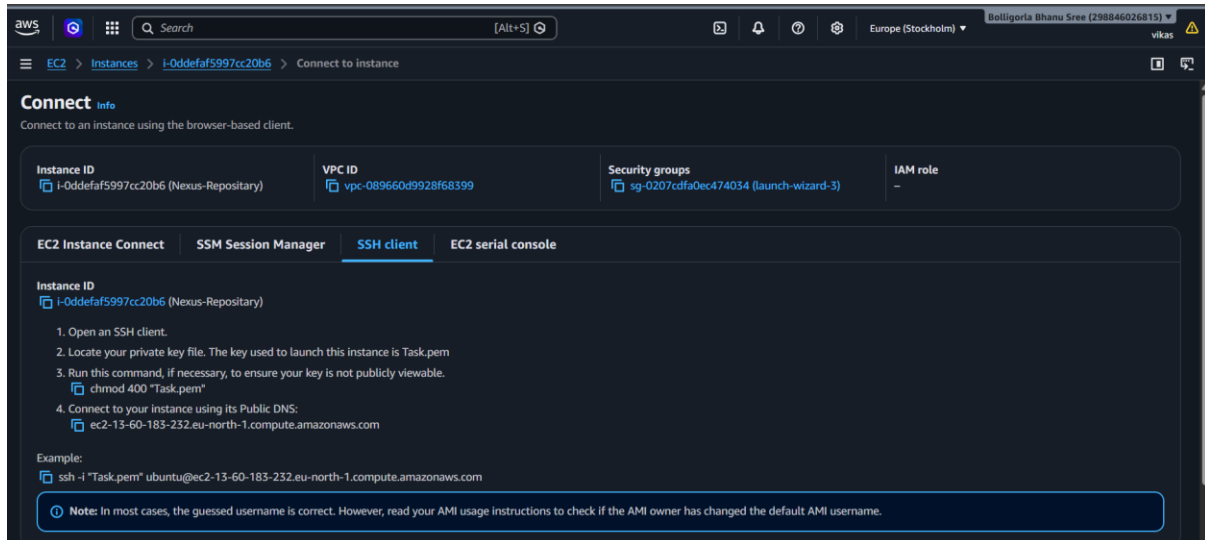
```
ubuntu@buildvm:~/Java-Web-Calculator-App$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.web.cal:webapp-add >-----
[INFO] Building WebAppCal Maven Webapp 0.0.3
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war-plugin-3.3.2.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war-plugin-3.3.2.pom (9.7 kB at 21 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/34/maven-plugins-34.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/34/maven-plugins-34.pom (11 kB at 202 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war-plugin-3.3.2.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.3.2/maven-war-plugin-3.3.2.jar (83 kB at 985 kB/s)
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ webapp-add ---
[INFO] skip non existing resourceDirectory /home/ubuntu/Java-Web-Calculator-App/src/main/resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ webapp-add ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ webapp-add ---
[INFO] skip non existing resourceDirectory /home/ubuntu/Java-Web-Calculator-App/src/test/resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ webapp-add ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ webapp-add ---
```

2. Nexus Repository Server Setup

Step 1: Install Java

`sudo apt update`

`sudo apt install openjdk-17-jdk -y`



```
ubuntu@Nexus-server:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@Nexus-server:~$ |
```

Step 2: Download Nexus

```
14 wget https://download.sonatype.com/nexus/3/nexus-3.85.0-03-linux-x86_64.tar.gz
ubuntu@nexusrepository:~$ cd nexus-3.85.0-03/
ubuntu@nexusrepository:~/nexus-3.85.0-03$
```

Step 3: Extract Nexus

`tar -xvf latest-unix.tar.gz`

Start Nexus

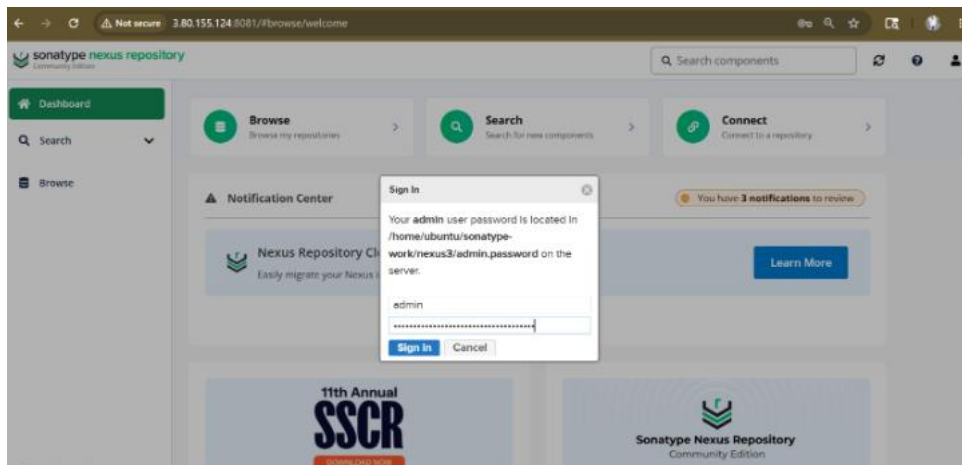
`cd /opt/nexus/bin`

`./nexus start`

```
16 tar -xvf nexus-3.85.0-03-linux-x86_64.tar.gz
17 ls
18 cd nexus-3.85.0-03/
19 ls
20 cd bin
21 ls
22 ./nexus
23 ./nexus start
24 ./nexus status
25 ./nexus status
```

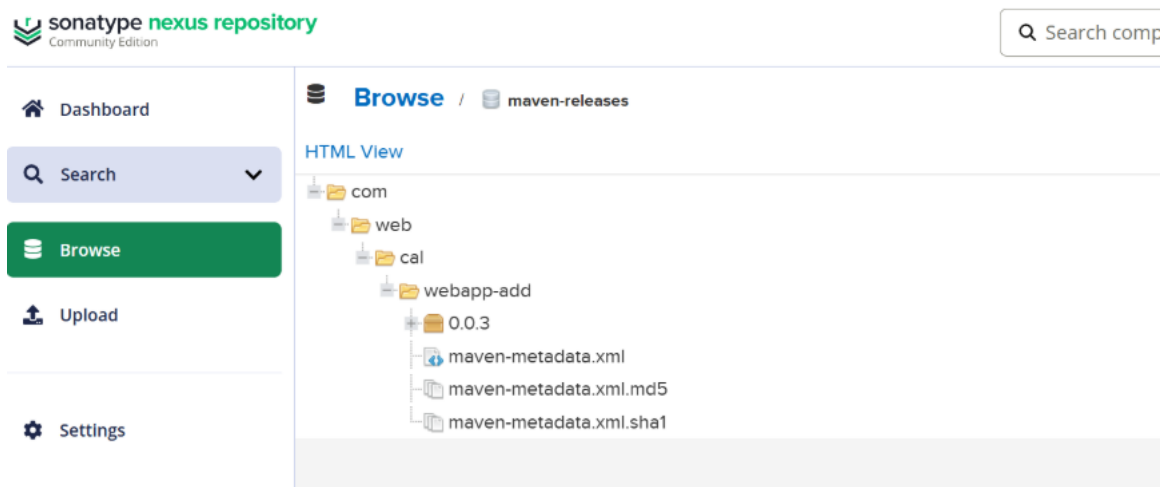
Step 5: Access Nexus Web Interface

<http://3.80.59.36:8081>



Step 6: Create Maven Hosted Repository

Create repository named maven-releases

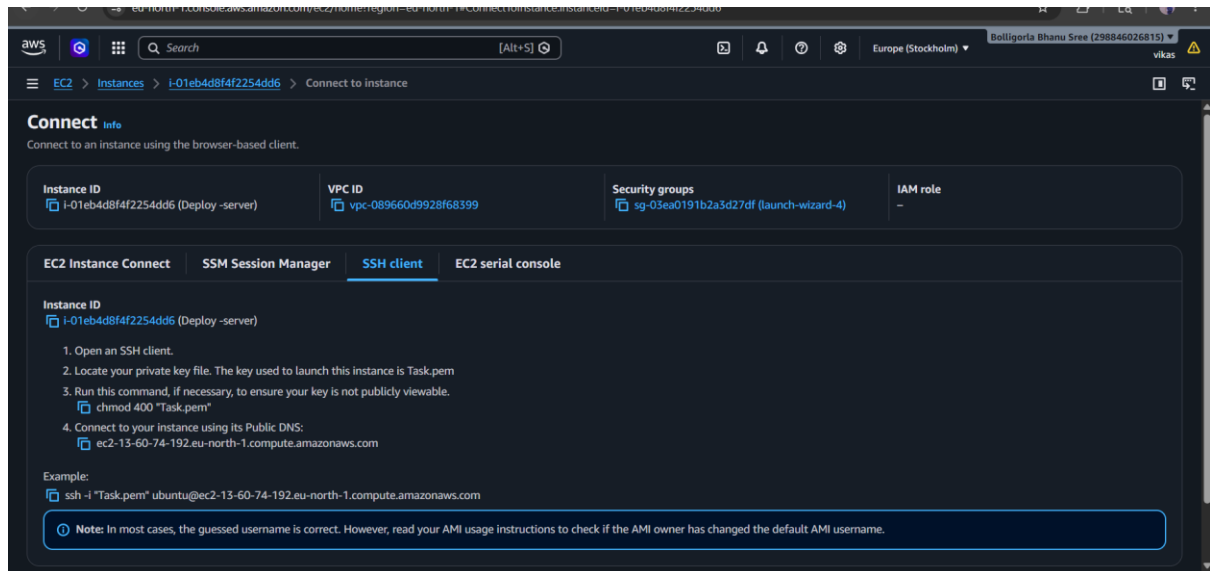


3. Deploy Server Setup

Step 1: Install Java and Apache Tomcat

`sudo apt update`

`sudo apt install openjdk-17-jdk -y`



```
ubuntu@Deploy-server:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@Deploy-server:~$ |
```

Download Tomcat:

`wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.tar.gz`

```
ubuntu@deployserver:~$ wget https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.zip
--2026-05-18 11:17:09-- https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.zip
Resolving d1cdn.apache.org (d1cdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to d1cdn.apache.org (d1cdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13642433 (13M) [application/zip]
Saving to: 'apache-tomcat-9.0.118.zip'

apache-tomcat-9.0.118.zip      100%[=====]
```

Extract Tomcat:

`tar -xvf apache-tomcat-9.0.118.tar.gz`

```
apache-tomcat-9.0.118  apache-tomcat-9.0.118.tar.gz
ubuntu@Deploy-server:~$
```

Step 2: Start Tomcat

cd /opt/tomcat/bin

./startup.sh

```
ubuntu@Deploy-server:~$ cd apache-tomcat-9.0.118/
ubuntu@Deploy-server:~/apache-tomcat-9.0.118$ ls
BUILDING.txt  LICENSE  README.md  RUNNING.txt  conf  logs  webapps
CONTRIBUTING.md  NOTICE  RELEASE-NOTES  bin  lib  temp  work
ubuntu@Deploy-server:~/apache-tomcat-9.0.118$ cd bin
ubuntu@Deploy-server:~/apache-tomcat-9.0.118/bin$ ls
bootstrap.jar  ciphers.sh  commons-daemon-native.tar.gz  daemon.sh  digest.bat  setclasspath.bat  startup.sh  tomcat-juli.jar  version.jar
catalina.bat  catalina.sh  commons-daemon.jar  configtest.bat  digest.sh  shutdown.bat  tomcat-native.tar.gz  tool-wrapper.bat  version.jar
ciphers.bat  configtest.sh  makebase.bat  startup.bat  tool-wrapper.sh
ubuntu@Deploy-server:~/apache-tomcat-9.0.118/bin$ ./startup.sh
Using CATALINA_BASE:   /home/ubuntu/apache-tomcat-9.0.118
Using CATALINA_HOME:   /home/ubuntu/apache-tomcat-9.0.118
Using CATALINA_TMPDIR: /home/ubuntu/apache-tomcat-9.0.118/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /home/ubuntu/apache-tomcat-9.0.118/bin/bootstrap.jar:/home/ubuntu/apache-tomcat-9.0.118/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

Not secure 54.166.218.21:8080

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Apache Tomcat/9.0.118

If you're seeing this, you've successfully installed Tomcat. Cool!



Recommended Reading:
[Security Considerations How-To](#)
[Manager Application How-To](#)
[Clustering/Session Replication How-To](#)

Developer Quick Start
[Tomcat Setup](#) [Realms & AAA](#) [Examples](#)
[First Web Application](#) [JDBC DataSources](#)

Step 3: Copy WAR File to Deploy Server

```
126 scp -i /home/ubuntu/devops.pem Java-Web-Calculator-App/target/webapp-add-0.0.3.war ubuntu@54.160.139.225:/home/ubuntu/apache-tomcat-9.0.118/webapps/
```

Addition

5

Calculator

first number:

Second number :

☒ addition

☐ subtraction

☐ production